



UiT The Arctic University of Norway

Faculty of Engineering Science and Technology

Electrification of truck transport between Hadsel and Narvik.

Elektrifisering av lastebiltransport mellom Hadsel og Narvik

Candidate numbers: 17, 19 and 15

INE-3800, Fall, 2025

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1 Introduction

The route between Hadsel and Narvik is a busy route, both for personal travel and freight transport. The route is the most used route for sea-food transport in Nordland (Hadsel kommune, 2022) which is Norway's second greatest sea-food county (Nofima, 2024). Ca. Therefore, the electrification of this route will represent a great cut in emissions and could as a pilot project also innovate and develop technology and infrastructure that could be used elsewhere.

On the way to a more sustainable future electrification of transportation, and its infrastructure, will be a very important step towards zero-emission goals. Norway's parliament has set a goal that all new trucks sold from 2030 and onwards should either be electric or run on biogas (Berge, 2023). A big part of passenger transport in Norway has transitioned towards electric over the last 10-15 years. But the heavy-duty freight has not. The challenge is obvious; the weight of vehicles and goods and its consequences makes electrification of freight transport harder compared to personal transport. The demand for battery size, charging speed, range and seasonal sensitivity is some of the reasons why freight transport has slower development in electrification.

This study will develop a mathematical optimization model to plan the electrification of the freight transport between Narvik and Hadsel. The model will be agnostic where battery capacity, range and charging are the parameters, and the results will later be compared to real truck models. The aim will be to optimize placement of the chargers and their capacity.

2 Theory

2.1 Route Information

2.1.1 Route Overview

The Hadsel Port to Narvik Port transport route is about 205 kilometres long and runs through the mountains, fjords and coastal areas of northern Norway. This route is a central part of the electric trucking network in Northern Norway, and its terrain and climatic conditions pose significant challenges for the range and energy consumption management of electric trucks.

Geographical location and route features:

The route starts at Hadsel Harbour (68.57°N, 14.91°E) and ends at Narvik Harbour (68.43°N, 17.43°E).

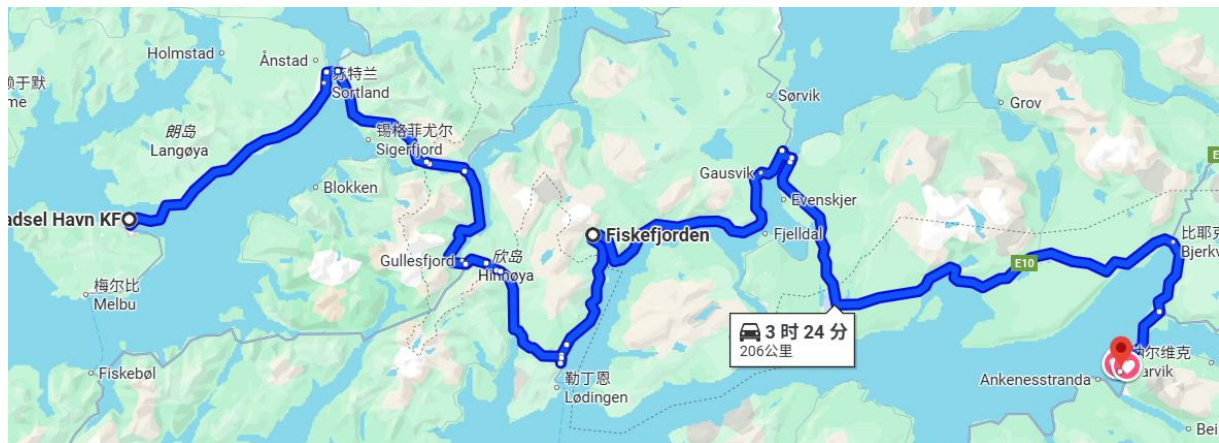


Figure 1 The entire route.

The route traverses the complex terrain of northern Norway and includes several uphill and downhill sections of the mountain. Uphill sections significantly increase energy consumption, while downhill sections may recover some of the energy through regenerative braking.

Sharp turns and narrow roads in the fjords and coastal zones place additional demands on the driver's operation and the management of the vehicle's energy consumption.

Climatic conditions:

Winter temperatures can be as low as -20°C , and snow and ice can cause slippery roads, increasing rolling resistance and braking distance.

Summer temperatures are mild (10°C to 20°C) and roads are in relatively good condition, but complex terrain still poses a challenge to transport efficiency.

Traffic Flow: According to the Norwegian Public Roads Authority (NPRA), the route is an important freight route in Northern Norway, connecting several ports and logistics centers. ((NPRA), 2023) Fixed-route freight trucks account for most of the traffic, while a small number of vehicles are long-distance transportation across the country.

2.1.2 Terrain Features

Analysis of terrain characteristics

The Hadsel Port to Narvik port route traverses the mountains, fjords and coastal zones of Northern Norway, and its complex terrain has a significant impact on the energy consumption and range of electric trucks.

Mountainous terrain: The route contains several uphill and downhill sections. Uphill sections can significantly increase the energy consumption of electric trucks, as the vehicle needs to overcome gravity to climb. Downhill sections may use regenerative braking technology to recover some of the energy to improve energy efficiency. According to research, regenerative braking recovers about 10%-30% of energy, depending on the slope and vehicle load (Cedillo-Campos, Pérez-González, Piña-Barcena, & Moreno-Quintero, 2019).

Fjord Terrain: Roads in the fjord area are usually narrow and come with sharp turns. This terrain limits the average speed of the vehicle and can lead to frequent acceleration and deceleration, which can increase energy consumption (Cherrett, et al., 2012). In addition, slippery roads in the fjord area, especially in winter, can further increase rolling resistance.

Coastal zones: Strong winds in coastal areas have an impact on the aerodynamics of electric trucks, especially crosswinds, which can increase the driving resistance of the vehicle, thereby increasing energy consumption (Liu, et al., 2022). Roads in coastal areas are usually flat, but salt spray can have a corrosive effect on the vehicle's battery and other electronic components.

The impact of terrain on energy consumption and endurance

Uphill sections: The additional energy consumption when going uphill can reduce the actual range of an electric truck by about 20%-30%, depending on the slope and vehicle load (Wang, et al., 2019).

Downhill sections: Regenerative braking technology can partially recover energy when going downhill, but the recovery efficiency is limited by the battery's charging capacity and slope length (Cedillo-Campos, Pérez-González, Piña-Barcena, & Moreno-Quintero, 2019)

Combined impact of complex terrain: Complex terrain (such as mountains and fjords) can result in an electric truck with an actual range of only 50%-70% of the theoretical value ((NPRA), 2023).

2.1.3 Road Conditions and Climate Factors

Road conditions and climatic conditions in Northern Norway have a significant impact on the operation of electric trucks, especially in extreme winter and summer conditions. Low temperatures, snow and ice in winter pose serious challenges to vehicle range and energy consumption management, while complex terrain still puts pressure on transportation efficiency despite the milder climate in summer.

Winter conditions have a particularly significant impact on the operation of electric trucks. Low temperature environments (e.g., -20°C) can significantly reduce the chemical reaction rate of the battery, resulting in a decrease in battery capacity and a possible 30%-50% reduction in the actual cruising range. In addition, auxiliary features such as cab heating further increase energy consumption, especially in extremely cold conditions. Snow and icy road conditions not only increase rolling resistance and braking distance, but can also lead to traffic delays, affecting transportation efficiency. Although road maintenance, such as snow removal and salting, has been able to improve traffic conditions, its effectiveness is limited, especially in the case of continuous snowfall. In addition, strong winds in winter adversely affect the aerodynamics of vehicles, and crosswinds can significantly increase driving resistance, further increasing energy consumption.

In contrast, mild climate conditions in summer are more favorable for electric truck operation. In the temperature range of 10°C to 20°C, the chemical reaction rate of the battery is close to

the optimal level, the energy loss is significantly reduced, and the actual cruising range can reach 70%-80% of the theoretical value. At the same time, roads are usually drier and flatter in summer, and rolling resistance is significantly reduced, which helps improve transportation efficiency. However, the complex terrain of mountains and fjords still poses challenges to energy consumption management. Despite mild summer temperatures in Northern Norway, persistent high temperatures can lead to battery overheating, affecting its charging efficiency and lifespan.

Overall, road conditions and climatic conditions in Northern Norway pose multifaceted challenges for electric truck operation. Low temperatures and harsh road conditions in winter significantly increase energy consumption and reduce range, while mild weather in summer contributes to transportation efficiency. However, whether in winter or summer, complex terrain and climate change remain key issues to overcome for electric truck operation. These factors need to be fully considered in the design and operational planning of electric trucks to ensure that they can meet the expected needs in actual operation in Northern Norway.

2.2 Truck Information

2.2.1 Truck Models and Technical Specifications

In the transportation environment of Northern Norway, electric trucks need to deal with complex terrain and extreme climatic conditions, so choosing the right model is crucial. Several electric truck models suitable for transportation in Northern Norway are listed below and their key technical parameters are summarized. These parameters include theoretical range, actual range (winter and summer), and battery capacity.

Table 1 Truck data parameters.

Brand/Model	Vehicle Type	Theoretical Range	Actual Range (Winter)	Actual Range (Summer)	Remarks
Volvo FL	16.7-ton box truck (urban distribution)	300 km			Battery capacity: 200-395 kWh
Volvo FE	27-ton/19-ton box truck	200 km			Battery capacity: 200-265 kWh
Volvo FM	44-ton/3-axle (regional transport)	380 km	100-150 km	150 km	Battery capacity: 180-540 kWh; significant range reduction in winter
Volvo FMX	44-ton (construction transport)	320 km			Battery capacity: 80-540 kWh
Volvo FH	44-ton (regional/urban transport)	300 km			Battery capacity: 180-540 kWh; sales started in October 2022
Scania L-series	29-ton (urban distribution, 2- or 3-axle)	250 km	14 miles (~22.5 km)		Significant range reduction in winter (-20°C)
Scania P-series	29-ton (regional distribution/construction driving)	250 km	14 miles (~22.5 km)		Significant range reduction in winter (-20°C)
Mercedes-Benz eActros	19-ton/27-ton (2- or 3-axle)	400 km			Battery capacity: 448 kWh; supports 160 kW DC fast charging

As can be seen from the table, there are significant differences in theoretical and actual range between different models. In winter, due to the impact of low temperatures on battery performance, the actual cruising range is usually only 30%-50% of the theoretical value. For example, the Volvo FM Electric has a theoretical range of 380 km, but in winter it is only 100-150 km in practice. In contrast, summer conditions are more favorable for battery performance, with actual range reaching 70%-80% of theoretical value.

In addition, battery capacity is an important factor in the range. The Mercedes-Benz eActros is equipped with a large 448 kWh battery for a practical range of up to 300 km in summer conditions, making it suitable for short and medium-distance transport. However, for routes that require long-distance transportation, it is still necessary to rely on the support of charging stations in the middle of the road.

2.2.2 Seasonal Impacts on Truck Performance

In winter conditions, particularly at low temperatures (e.g., -20°C), the performance of electric vehicle batteries is significantly reduced, resulting in a substantial decrease in the actual driving range compared to theoretical estimates. Low temperatures slow down the

chemical reactions within the battery, thereby reducing the available energy. Additionally, auxiliary functions such as cabin heating further increase energy consumption, which is particularly pronounced under extreme cold conditions. According to project test data, the Volvo FM, with a theoretical range of 380 km, achieves only 100-150 km in winter, representing 30%-50% of its theoretical range. Furthermore, road conditions such as snow and ice increase rolling resistance, exacerbating energy consumption.

In summer conditions, characterized by moderate temperatures (10°C to 20°C), battery performance improves significantly, leading to an increase in driving range compared to winter. Under these conditions, the chemical reactions within the battery operate near optimal efficiency, and energy losses are minimized. Project tests indicate that the Volvo FM achieves an actual range of up to 150 km in summer, approximately 70%-80% of its theoretical range. However, factors such as terrain (e.g., mountainous and fjord regions) and payload weight must still be considered, as they can contribute to additional energy consumption and range variability.

2.3 Current Charging Infrastructure

2.3.1 Charging Station Distribution

The transport route from Hadsel Havn KF to Narvik in Northern Norway is about 206 kilometers long and runs through mountains, fjords and coastal areas. The distribution of charging infrastructure along the route is critical to support the operation of electric trucks. The layout of existing charging stations is mainly concentrated in three key nodes: starting point, middle way and ending point, basically covering the main needs of transportation routes.

The starting point charging station is located in the port of Hadsel Havn KF, which serves as the starting point of the transportation route, providing a fully charged state for the truck, ensuring that it can start its transportation tasks smoothly. The charging stations at Hadsel Port are strategic, providing power support not only for the initial transportation of trucks but also for the planning of endurance for subsequent transport.

Midway charging stations are located in the Sortland and Gullesfjord areas. These charging stations are important nodes in the transportation route, providing fast charging services for trucks transported over long distances, alleviating the problem of insufficient range. The

Sortland charging station is located at the front of the route and is suitable for providing replenishment at the beginning of transportation, while the Gullefjord charging station is located in the middle of the route and is a key resupply point that can support the truck's endurance needs after complex terrain and energy-intensive sections.

The terminal charging station is located in the port of Narvik and provides charging support for trucks completing transport tasks and ensuring their endurance on the return journey. The charging stations at Narvik Port are well-laid out to meet the charging needs of trucks after completing their tasks while preparing them for the next shipment.

From the perspective of map analysis, the distribution of existing charging stations basically covers the key nodes of the transportation route (starting point, middle way, and ending point). The setting of midway charging stations can effectively alleviate the problem of insufficient battery life caused by complex terrain and high-energy consumption sections in long-distance transportation. However, some sections may require further optimization of the distribution of charging stations to ensure endurance in adverse weather conditions.

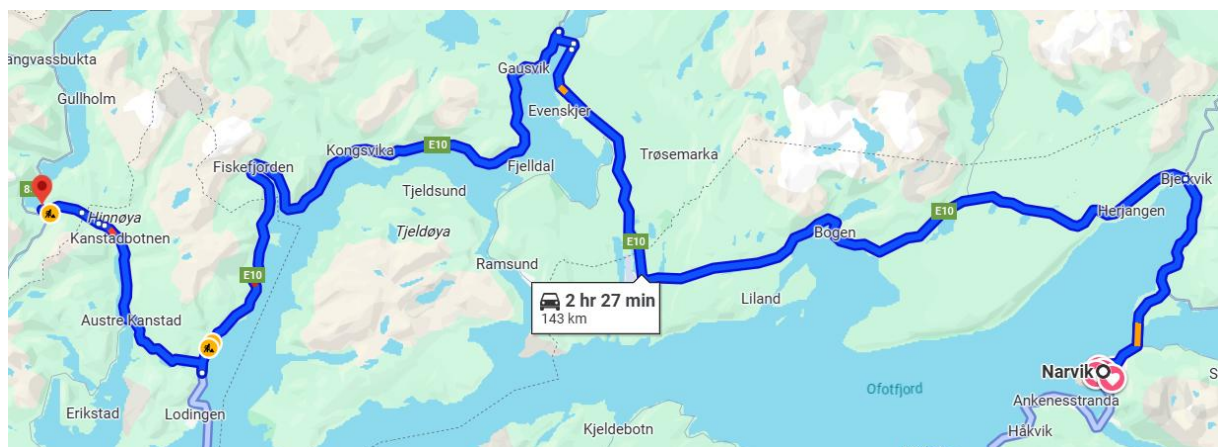


Figure 2 Hadsel Port to Narvik port transport route

The distribution of existing charging stations can basically meet the charging needs of electric trucks on the Hadsel Port to Narvik port transport route. However, to further optimize transportation efficiency, it is recommended to add a stopover charging station between Gullefjord and Narvik to cope with the impact of complex terrain and extreme weather on the range.

2.3.2 Technical Capabilities and Capacity

Charging infrastructure in Northern Norway has certain technical advantages in supporting electric truck operations, but there are still shortcomings in coverage and charging speed. The application ratio of fast and slow charging technologies directly affects transportation efficiency, while the limitations of existing infrastructure pose challenges to the long-distance transportation of electric trucks.

The technical level of existing charging stations is mainly reflected in the application of fast charging and slow charging technology. The starting point charging station is located in the port of Hadsel and is equipped with slow charging equipment for long periods of time before the truck departs, ensuring that the vehicle starts its transport mission with a full charge. Medium-way charging stations are located in Sortland and Gullsfjord, both equipped with high-power fast charging equipment, capable of charging trucks to 80% in 30-45 minutes, significantly improving transportation efficiency. The terminal charging station is located in the port of Narvik and supports the charging needs after completing the transport mission and provides endurance for the return journey. In addition, the long-distance truck stop in Lødingen offers slow charging services for nighttime or long stops, helping to reduce the load on the grid.

Although existing charging stations are technically capable of meeting basic needs, there are still some shortcomings. Firstly, the coverage of charging stations is limited, especially in the section between Gullsfjord and Narvik, where there is a lack of support for midway charging stations. This deficiency can lead to the risk of insufficient range for electric trucks in adverse weather conditions. Secondly, existing fast charging devices typically have a power range of 150-250 kW, and there is still room for improvement in charging efficiency compared to the latest ultra-fast charging technologies (such as devices above 350 kW) [3]. In addition, grid capacity constraints are one of the main challenges facing current infrastructure. Remote areas in northern Norway have weak grid coverage, making it difficult to support the widespread deployment of high-power charging equipment, especially during peak winter electricity usage.

Table 2 Candidate charging stations.

No.	Location	Analysis
1	Eviny Charging Station (Hadsel)	Located at the starting point of the route, suitable as an initial charging station to provide trucks with a full charge.
2	Circle K (Sortland)	Situated in the Sortland area, an important mid-route charging node suitable for short-duration fast charging.
3	Eviny Charging Station (Gullesfjord)	A critical mid-route node, suitable for providing fast-charging services for long-haul trucks.
4	Lødingen Long-Haul Truck Stop (Overnight)	Provides overnight charging services, suitable for trucks with extended stops, supporting slow charging to reduce grid load.
5	Evenes Circle k	Provide a charging station with gas station facilities.
6	Narvik Charging Station	Located at the endpoint of the route, supports trucks completing their tasks and ensures sufficient range for return trips.

Through the specific analysis of the candidate charging stations, it can be found that the functional distribution of the existing charging stations is reasonable, but it still needs to be further optimized. The fast charging stations in Sortland and Gullesfjord can effectively alleviate the pressure of long-distance transportation, while the slow charging stations in Lødingen support long-term stops. However, the section between Gullesfjord and Narvik lacks charging station coverage, and it is recommended to add a fast charging station in this area to further improve transportation efficiency and safety.

Northern Norway's charging infrastructure already has a certain foundation in terms of technical level and capacity, but it still needs to be optimized in terms of coverage, charging speed and grid capacity. Future development directions include the introduction of higher-power ultra-fast charging equipment, increased grid capacity, and the addition of charging station layouts in key road sections to meet the operational needs of electric trucks in complex terrain and extreme climatic conditions.

2.4 Summary

The actual conditions in Northern Norway pose multifaceted challenges for the operation of electric trucks, while also providing clear directions for the optimization of charging infrastructure. The complex terrain and extreme climatic conditions significantly impact the energy consumption and range of electric trucks, while the inadequacies of existing charging infrastructure further limit their operational efficiency.

The terrain of Northern Norway is dominated by mountains, fjords and coastal areas, with several uphill and downhill sections in the transport route. Uphill sections significantly increase vehicle energy consumption, while downhill sections can partially recover energy through regenerative braking, but overall energy consumption is still high. In addition, low temperatures, snow and strong winds in winter adversely affect battery performance and vehicle aerodynamics, resulting in actual range often only 30%-50% of theoretical values. While mild climate conditions in summer improve battery performance, complex terrain still poses challenges to transportation efficiency. Together, these factors suggest that the topography and climate of Northern Norway pose significant challenges to the energy consumption management of electric trucks.

The existing charging infrastructure already has a certain foundation in supporting the operation of electric trucks, but there is still a lack of coverage and technical level. The distribution of charging stations is mainly concentrated at the starting point (Hadsel Havn KF), the midway point (Sortland and Gullesfjord), and the end point (Narvik), capable of meeting basic transportation needs. However, the lack of charging station coverage on the section between Gullesfjord and Narvik can put vehicles at risk of insufficient range in adverse weather conditions. In addition, the power of existing fast charging equipment is usually between 150-250 kW, which is still room for improvement compared to the latest ultra-fast charging technology, and the capacity constraints of the grid in remote areas also pose challenges to the deployment of high-power charging equipment.

The analysis in this section provides important theoretical support for the subsequent chapters (Method and Results). Through a detailed study of the topography, climatic conditions and current status of charging infrastructure in northern Norway, the main problems faced in the operation of electric trucks, and the optimization direction are identified. These theoretical foundations will provide data support for subsequent mathematical modeling and result analysis and provide a decision-making basis for the optimization of charging station site selection and technical upgrades.

3 Method

3.1 AMPL model

This chapter will explain the mathematical programming model that was used in Ampl. The philosophy of the model is to make all the proposed charging places nodes in a set N . The decision variables $v(N)$ is a binary value that decides if we build a charging station there or not, and $p(N)$ decides how many plugs there should be at each station. The objective function of the model is to optimize, or minimize to be more specific, the number of charging stations while fulfilling the needs of the electric trucks.

3.2 Mathematical programming model

To simulate the route between Hadsel and Narvik a mathematical version of the model was made in AMPL.

3.2.1 Assumptions

This section will explain the assumptions we have made in the model. At first some general assumptions are made clear, then some specific ones are mentioned.

The general assumptions are that energy use will be deterministic, and one set value for the whole route. For the winter scenario a scaling will be added onto the assumed summer energy use. The model assumes one corridor between the places, with no sideroads or detours. For charging time, the speed of charge will be linear even though most vehicles charge slower at the start and end of the battery cycle, typical values are slow charging between 0-10% state of charge and 80-100% state of charge. This way a parameter of charging time could be equal throughout the model.

Nodes and geography

The nodes in the mathematical model will be the different locations where a possible charging station could be. In this instance we have six different locations where we will assess locating a charging station, these are: Hadsel, Gullesfjord, Lødingen, Evenes, Bjerkvik and Narvik. With these locations the longest part of the route without a charger is 70km.

Customers

The electric trucks could be used for freight transport from end to end (Hadsel-Narvik) but could also be used to go between different places in between these places. For example, Rema 1000 which is one of the biggest grocery chains in Norway, have its distribution center for northern Norway in Bjerkvik.

Model choice

The model will be model agnostic, meaning that the output and goal of the study is to find out which charging station that will be opened, and then find the range between them. Then the results from the model will be used to determine if today's market could supply trucks that fit this new model. This way the model does not have to include complicated coding where different trucks are introduced in the code, but one could instead assess the output of the model and if it fits the capacity.

Scenario

This model's scenario is based on a worst-case scenario. It assumes that the truck chosen for the mission is the Volvo FM with 100km range in winter. By assuming winter and worst case scenario, the result will be robust and give redundancy in summer and in case of improvement of range or higher traffic demands.

3.2.2 Parameters

The four parameters that are crucial for the model is the position of the nodes, assumed worst case range for truck and the minimum number of plugs on starting nodes and nodes in the middle of the route.

Hadsel is decided to be the “zero point” of distance, then google maps is used to find the distance from Hadsel to the nodes. The model finds the distance between nodes by using position of the current node then subtracts the distance to the last node.

The range parameter simulates the range of the electric trucks. In the model it is used so that no stretch in the route without a fast charger is longer than the trucks range R .

For deciding the number of plugs at each point, both end nodes and middle points, we have used the assumed traffic to decide a number of plugs needed in the stations. This is not calculated in the model but set by the model in the data file.

3.2.3 Variables

The variables are one binary decision variable that decides if we should build one or not, and the other is how many plugs, that are ultimately decided by the parameters. But the variables choose which station they should be at.

3.2.4 Restrictions

In successive order in the model the restrictions are first two that “forces” the start and end nodes to have a charging station built at the locations. This is an assumption since the project aim is to electrify transport between these nodes, therefore it is a certainty that these should have chargers built.

The next constraint is the minimum plugs constraint that state the minimum plugs at each node. It is made “over” the set N so that you could define minimum number of plugs at each station. In this model we used it to define number of plugs in the start and end nodes.

The plugs at built charging stations is to ensure that the number of plugs at a station is equal to or bigger than the binary decision value. That ensures that a charging station is not built without plugs.

The middle station capacity is a constraint that makes the middle stations have at least the stated minimum number of plugs that were defined in the parameter that defines the least number of plugs at a middle station.

The Range constraint calculates the distance between nodes as mentioned earlier and ensures that the maximum distance between to charging stations Is always less than the range expressed in parameter R.

3.3 Method for finding assumed traffic numbers

To suggest placement and number of electric truck chargers on the route between Narvik and Hadsel, it is important to find an assumption for the number of heavy trucks going this route. To find an assumption data was gathered from Statens Vegvesen to see the number of heavy trucks that crossed different trackers that they have put out at strategic places (Statens vegvesen, u.d.). These strategic places are mostly before and after every crossroad on the main roads. That way one could see every input and every output of the crossroad. Then the

plan was to do algebraic procedure to find the total amount of heavy trucks going from Hadsel to Narvik.

After gathering the data from the official website, it was quickly realized that there is too much uncertainty about using this data and the best way to approximate the heavy traffic was to do ballpark measures. There were many reasons for this, but the main reason was that many of the important traffic measurement stations were not in working order. For instance, the measurement stations for the road coming from Lødingen, where the ferry Bognes – Lødingen comes from which is an often used ferry for freight transport, were not operational. The same situation was at another very important intersection for freight transport, the crossroads from Harstad, that is a city bigger than Hadsel, therefore one could conclude that, only on this basis, the data has too much uncertainty. One could also argue that the “shadow numbers” from freight transport going short distances back and forth, inside the cities, also add additional uncertainty to these numbers.

The best assumption will be to use a procedure where the maximum limit would be equal to the average number of heavy-duty traffic that leaves Hadsel. Then we use a share of that number that we assume is the number of transports daily. The daily average over 2024 was 240 heavy trucks going out from Hadsel, direction towards Sortland/Narvik each day (Statens vegvesen, u.d.). Considering that this number represents every truck going from Hadsel, one could argue that only a small fraction of these will have Narvik as the end location, even though we have stated before that this is an often used route. To have some kind assumption of working with we assume that 24 trucks (10%), go from Hadsel to Narvik every day of the year.

3.3.1 Effects on the model

The Assumed amount of traffic through this route affects how many chargers that are chosen to have at each point. The model is robust, and any number could be inserted into it. The assumption made that 24 trucks go from Hadsel to Narvik, gives naturally that 24 trucks also go the other way.

This affects the number of plugs at the starting nodes, and the middle nodes. Since every starting point should give all the trucks starting from those the possibility to start the journey with full state of charge, it is stated in the model that the number of chargers should be half of the number of total trucks going from the starting nodes. For example, with one hour charging

speed and the number of plugs at starting nodes equal half of the number of trucks, then all trucks could be charged over a timespan of two hours. This is assumed to be feasible for this model.

For the middle points one would have to choose how many plugs you want. It is important to take into consideration that there is traffic going both ways that are in need of the charging stations, and they could meet in the middle point.

4 Results, discussion and conclusion.

Results obtained by the AMPL model show that to have optimal placement of the charging stations there should be built one charging station at the start and end node, then one station in Gullefjord and one in Evenes, both with 16 plugs. This results in total 4 charging stations and 56 plugs in total. The results are obtained from AMPL by using the files from the appendix, electric trucks model file, data file and run file.

```
ampl: include trucks.run;
CPLEX 22.1.2: 0000000000000000CPLEX 22.1.2: optimal solution; objective 4
0 simplex iterations
v[1] = 1
p[1] = 12

v[2] = 1
p[2] = 16

v[3] = 0
p[3] = 0

v[4] = 1
p[4] = 16

v[5] = 0
p[5] = 0

v[6] = 1
p[6] = 12
```

Figure 3 snippet from console in AMPL that display the results from the mathematical model.

The result could be read as nodes 1 – 6 which are the nodes in order from Hadsel to Narvik in order. Parameter v, as stated, is the decision variable, and p displays the number of plugs at the given location.

4.1 Discussion

The results obtained could be discussed from the point of uncertainty about the traffic. Since the traffic number were not possible to obtain the model were made to be flexible in the sense that the amount of traffic could be taken into account by adding more or less charging plugs, and the result given is only for one case where the amount of traffic assumed is 24 trucks starting from the start and end node.

What is certain is that, with the suggested placement of the charging station(nodes), there will be a charging station available for trucks that have a range of 100km or more to drive from

Hadsel to Narvik. They will be able to charge at the start and finish of the route and have the opportunity to charge along the way if the road conditions, weather conditions, range, or load force them to do so.

4.2 Conclusion

The conclusion is that according to the model made in this report the finished charging infrastructure for the “green corridor” project should be 4 charging stations at Hadsel, Gullsfjord, Evenes and Narvik. Hadsel and Narvik should have 12 plugs each to accommodate the trucks that start from these nodes, and the middle point chargers should have 16 chargers each to accommodate crossing traffic.

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Appendix

Declaration of generative AI and AI-assisted tools in the writing process

During the preparation of this work the authors used the generative AI solution recommended by UiT in order to:

- Brainstorm ideas during the preparation.
- Make a disposition.
- Find sources of information.
- Help write the code, make constraints and defining parameters to make the code work.

After using this tool/service, the author(s) reviewed and edited the content as needed and takes full responsibility for the content of the report.

Analysis of the traffic

The table below shows “cleaned” up data for the passing of heavy trucks longer than 5,6m on selected crossroads and intersections. Length quality gives a measure of confidence for length measurement.

cross no.	road explanation		Year	length quality (%)	Lane	Total heavy vehicles
cross 1	main road from Stokmarknes into cross 1	Skagen	2024	99,7	Total in direction Sortland	240
	side road into cross 1	Sortland north	2024	99,7	Total in direction Sortland	311
	side road out of cross 1	Sortland north	2024	99,7	Total in direction Frøskeland	307
	side road into cross 1	Strand north	2024	99,5	Total in direction Sortland	194
	side road out of cross 1	Strand north	2024	99,6	Total in direction Andenes	228
cross 2 (all data available)	main road out of cross 1	Strand	2024	99,9	Total in direction Gullefjordbotn	421
	side road out of cross 1	Strand	2024	99,9	Total in direction Sortland	500
	side road into cross 3	E10 Gullefjord south	2024	99,5	Total in direction Narvik (Bjerkvik)	189
	side road out of cross 3	E10 Gullefjord south	2024	99,8	Total in direction Svolvær	199
	main road into cross 3	Gullefjord west	2024	99,9	Total in direction Gullefjordbotn	349
cross 3 (missing data on 2 side roads)	side road out of cross 3	Gullefjord west	2024	99,9	Total in direction Sortland	398
	main road out of cross 3	E10 Gullefjord east	2024	99,5	Total in direction Narvik (Bjerkvik)	349
	side road out of cross 3	E10 Gullefjord east	2024	99,4	Total in direction Svolvær	429
	side road out of cross 4	Kåringen north	2024	99,6	Total in direction Narvik (Bjerkvik)	264
	main road out of cross 4	Kåringen north	2024	99,5	Total in direction Svolvær	206
cross 4 (missing data on 2 side roads)	main road out of cross 5	Tjeldsund bridge west	2024	99,8	Total in direction Narvik (Bjerkvik)	584
	side road out of cross 5	Tjeldsund bridge west	2024	100	Total in direction Svolvær	725
	main road into cross 6	Bjerkvik west	2024	99,9	Total in direction Narvik (Bjerkvik)	431
	side road out of cross 6	Bjerkvik west	2024	99,9	Total in direction Svolvær	413
	end (number of vehicles to Narvik)	Bjerkvik	2024	100	Total in direction Narvik	564
cross 5 (all data available)	side road out of cross 6	Bjerkvik	2024	100	Total in direction Nordkjosbotn	550
	side road into cross 6	Øse	2024	99,9	Total in direction Narvik	463
	side road out of cross 6	Øse	2024	99,6	Total in direction Nordkjosbotn	446

AMPL model, including model, data and run file

.mod file:

```
set N ordered;          # Nodes in route order (1..m)

param pos{N} >= 0;       # Position along route (km), increasing in data
param R > 0;             # Max travel distance between stations (range)
param min_plugs{N} >= 0 default 0; # Minimum plugs per node (e.g., depots)
param mid_min_plugs integer >= 0 default 1; # Peak plugs at en-route stations
param s symbolic := first(N); # Start node (default: first)
param t symbolic := last(N);  # End node (default: last)

var v{N} binary;        # 1 if a station is built at node i
var p{N} integer >= 0;  # Number of plugs at node i

minimize TotalStations: #minimizing the number of stations built to accomodate trucks
    sum{i in N} v[i];

subject to StartActive: v[s] = 1;
subject to EndActive:   v[t] = 1;

#stating thew number of plugs at stations and at the start/end nodes.
subject to PlugMinimum{i in N}: p[i] >= min_plugs[i];
subject to PlugAtBuilt{i in N}: p[i] >= v[i];

subject to MidStationsCapacity{i in N diff {s,t}}: #number of plugs in the middle stations that stated/chosen in the .dat file
    p[i] >= mid_min_plugs * v[i];

subject to RangeChain{j in N: j != first(N)}: #Range constraint making sure that the distance between every chargin station is less than range
    sum{i in N: ord(i) < ord(j) and pos[j] - pos[i] <= R} v[i] >= v[j];
```

.dat file:

```
set N := 1 2 3 4 5 6;      # 1=Hadsel, 6=Narvik

param pos := #in successive order; Hadsel, gullefjord, lodingen, evenes, bjerkevik, narvik.
1 0
2 63
3 84.5
4 144
5 188
6 220 ;

param R := 100;

# Depot minimum plugs
param min_plugs :=
1 12
2 0
3 0
4 0
5 0
6 12 ;

# Peak requirement at route stations (simultaneous arrivals)
param mid_min_plugs := 16;
```

.run file:

```
reset;  
model Electrictrucks.mod;  
data trucks.dat;  
  
option solver cplex;  
  
solve;  
  
display {i in N}: v[i], p[i];
```